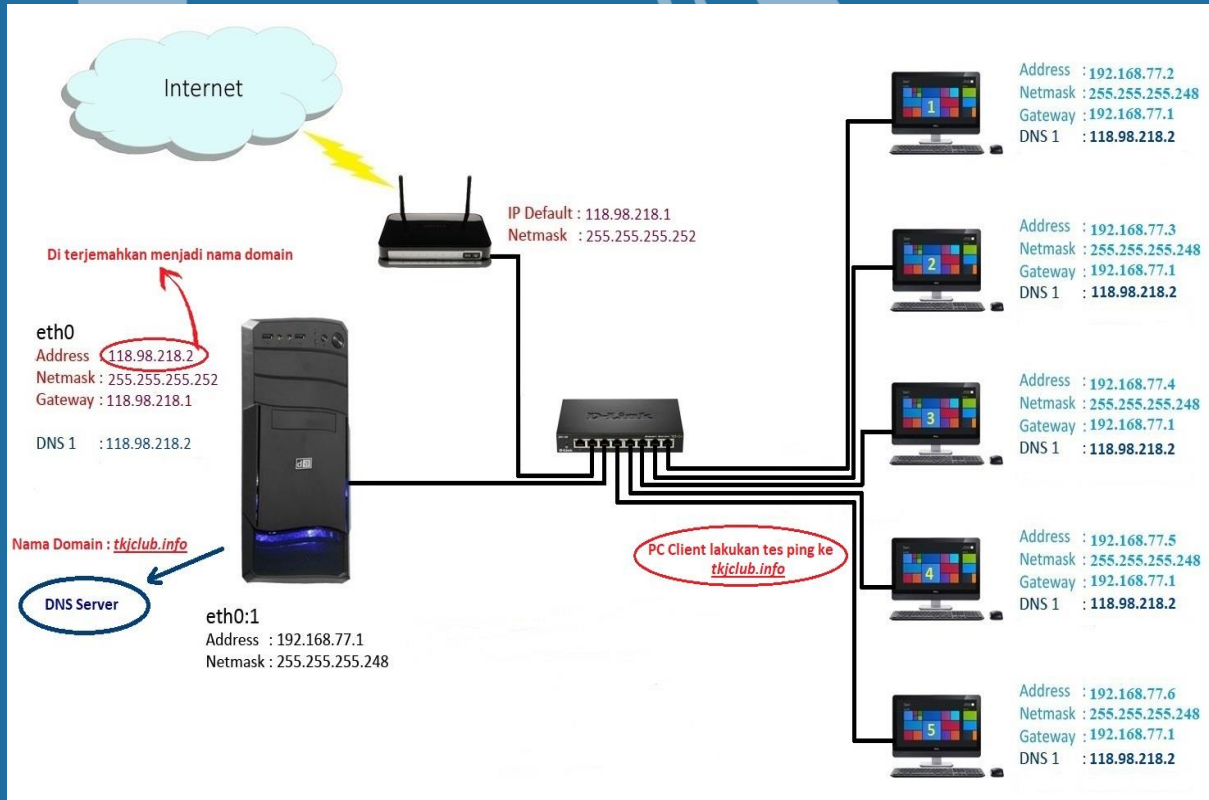


DNS SERVER



SALAM SUKSES TKJ CLUB

<http://www.tkjclub.net/tkj>



debian 10

Created by : M. Asriadi

A. Definisi DNS Server

DNS Server (Domain Name System) adalah Server yang ditujukan untuk menerjemahkan alamat IP menjadi nama domain begitupun sebaliknya.

B. Fungsi DNS Server

- Untuk menerjemahkan alamat IP menjadi nama domain, begitupun sebaliknya.
- Agar client lebih mudah mengakses suatu alamat.

Dengan adanya konfigurasi **DNS** ini kita lebih mudah mengakses Server tanpa harus menghafal IP Address, kita hanya perlu mengingat nama domain karena pada dasarnya kebanyakan orang akan lebih mudah mengingat nama dibandingkan angka.

Contoh :

google.com merupakan contoh nama domain yang diterjemahkan ke IP 8.8.8.8 atau 8.8.4.4 (IP Public Google). Melakukan test PING ke **google.com** sama saja melakukan test PING ke 8.8.8.8 atau 8.8.4.4 . Akan tetapi nama **www.google.com** lebih mudah diingat dibandingkan IP 8.8.8.8 atau 8.8.4.4 .

C. Script

```
>>> Pastikan konfigurasi Routing telah berhasil
# apt-cdrom add
>> masukkan DVD-2
# apt-get update
# apt-get install -y bind9
# apt --fix-broken install
# apt-get install -y dnsutils
# cd /etc/bind
# nano named.conf.local
```

>>>Tambahkan script :

```
zone      "namadomain"{
    type master;
    file "/etc/bind/db.name";
};

zone      "3_oktet_dibalik.in-addr.arpa"{
    type master;
    file "/etc/bind/db.ip";
};
```

Save dengan Ctrl + X >> tekan Y >> tekan Enter

```
# cp db.local db.name
# nano db.name
```

>>> Edit Script

```

                                namadomain. root.namadomain.
                                _____
                                _____
                                _____
                                _____

;
      NS      hostname.namadomain.

;
      A      IP Address
hostname  A      IP Address

```

Save dengan Ctrl + X >> tekan Y >> tekan Enter

```
# cp db.127 db.ip
# nano db.ip
>>>Edit Script
```

```

                                namadomain. root.namadomain.
                                _____
                                _____
                                _____
                                _____

;
      NS      hostname.namadomain.

;
0ktet 4  PTR      namadomain.
0ktet 4  PTR      hostname.namadomain.

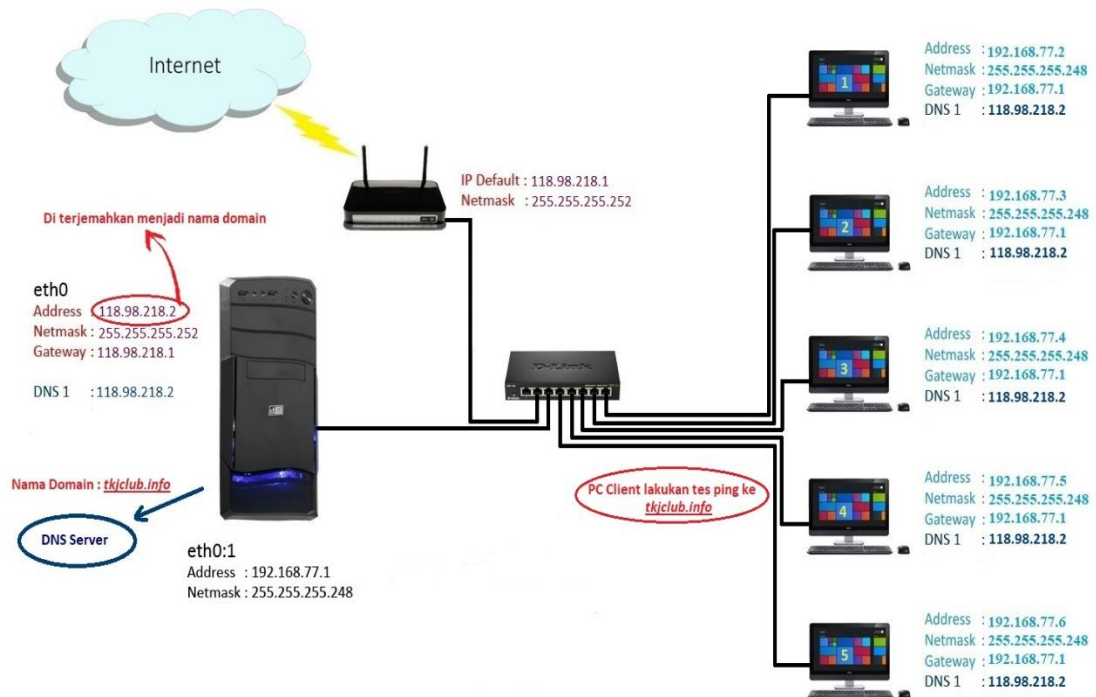
```

Save dengan Ctrl + X >> tekan Y >> tekan Enter

```
# /etc/init.d/bind9 restart
# nslookup namadomain
# nslookup IP Address
# ping namadomain
```

} Di Server
Di CMD

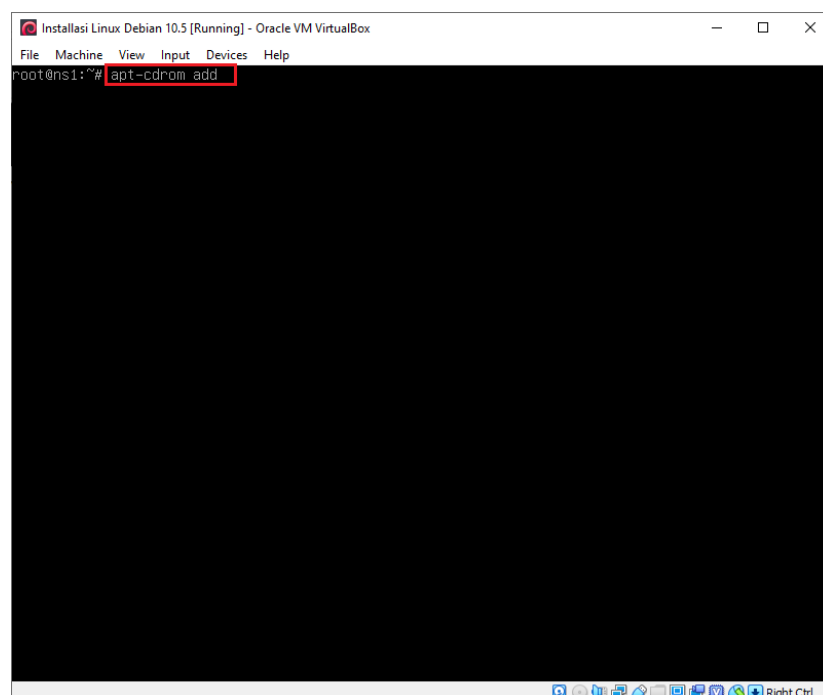
D. Topologi



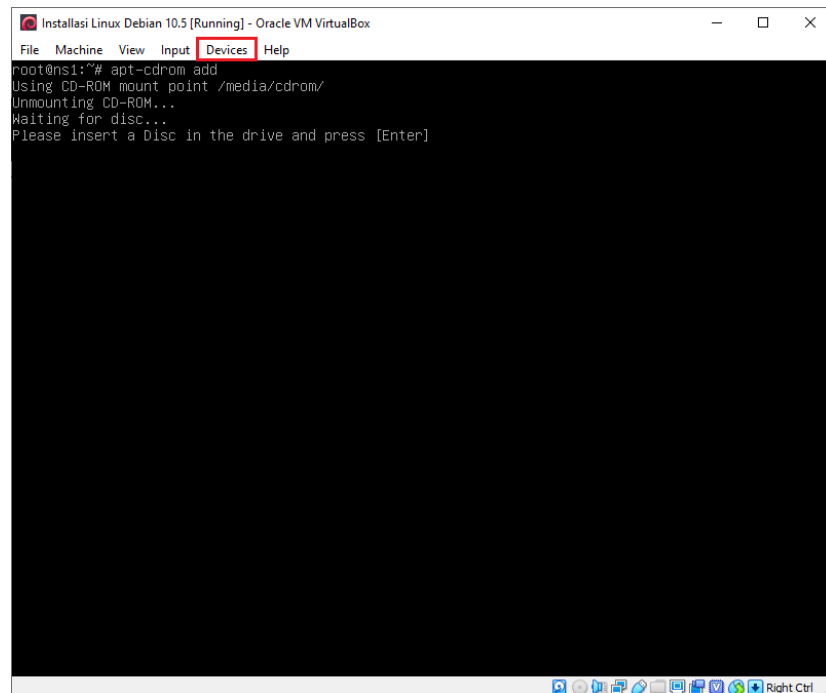
E. Langkah-Langkah Konfigurasi DNS Server

1. Pastikan konfigurasi Router Server telah selesai dan berhasil.
2. Masukkan atau tambahkan DVD-2 Debian karena paket aplikasi DNS Server ada di DVD-2. Ketikkan perintah berikut :

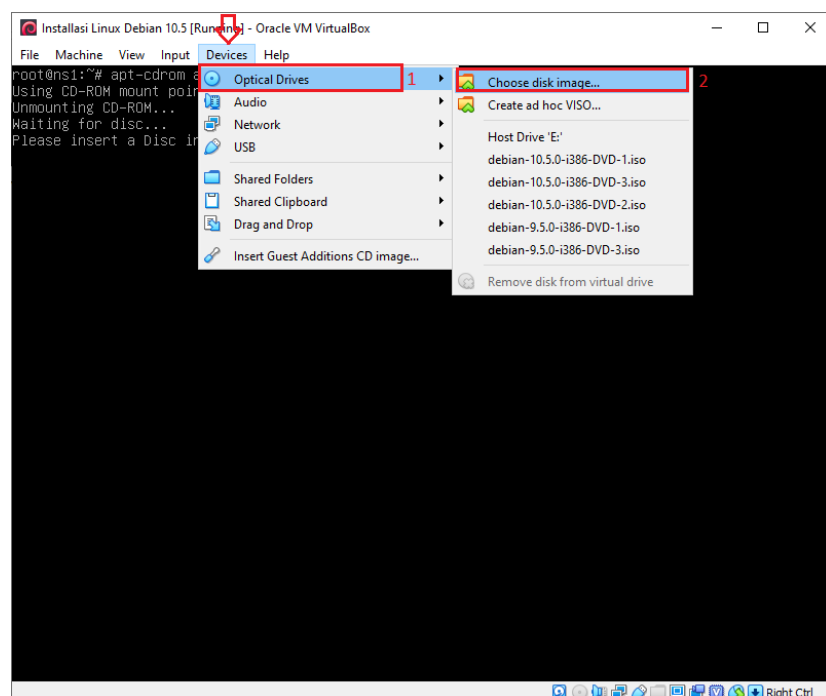
apt-cdrom add >>lalu tekan Enter



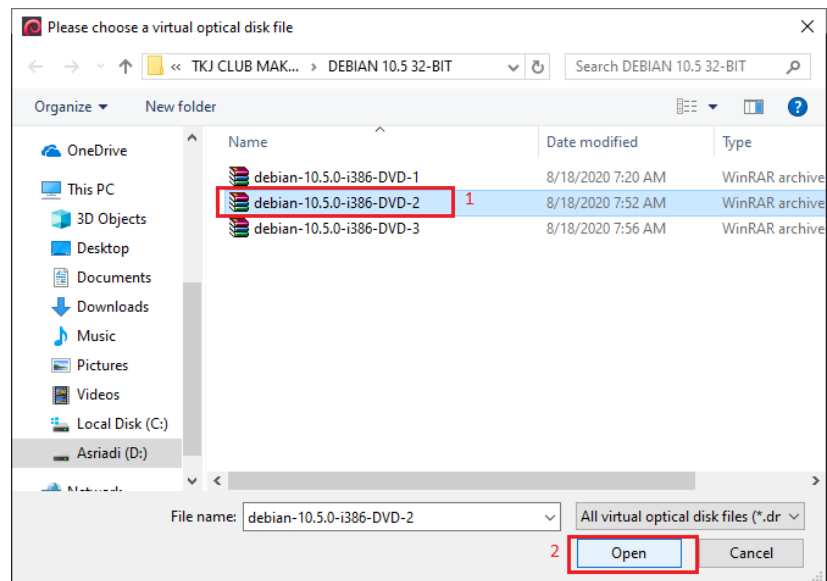
3. Jika tampil seperti gambar berikut. Di ingatkan jangan menekan Enter dulu. Pertama klik **Devices**.



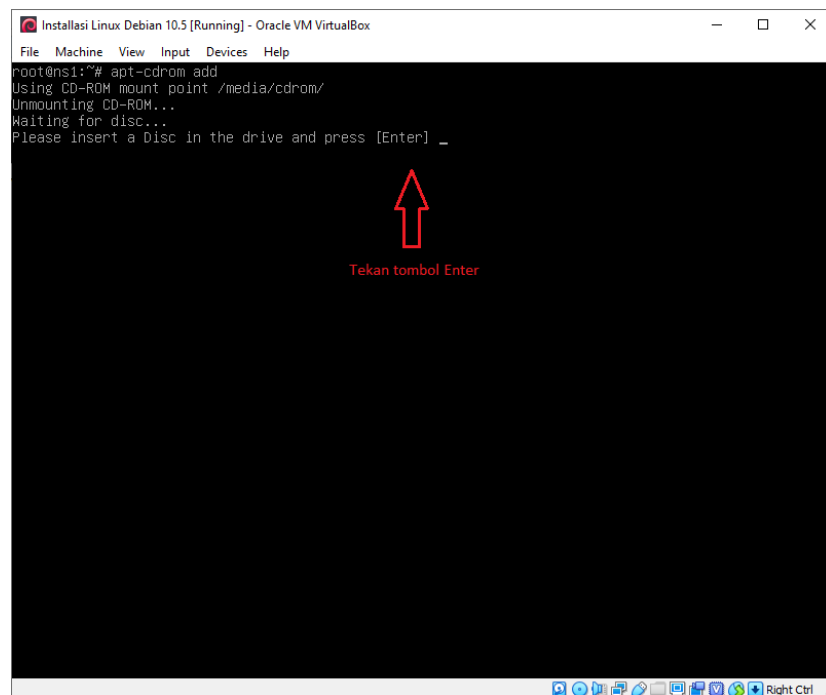
4. Klik **Devices >> Optical Drives >> Choose disk image...**



5. Lalu pilih **debian-10.5.0-i386-DVD-2**, lalu klik **Open**.



6. Kemudian tekan **Enter**.



7. Jika tampilannya seperti gambar di bawah, berarti DVD-2 telah berhasil di tambahkan.

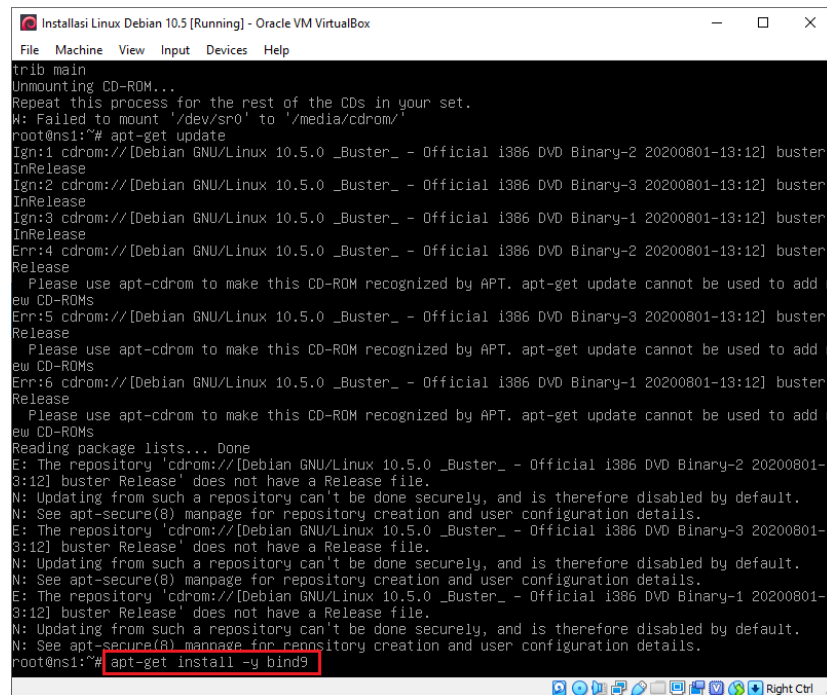
8. Setelah memasukkan DVD, selanjutnya update. Masukkan perintah berikut :

apt-get update >>lalu tekan *Enter*


http://www.tkjclub.net

9. Selanjutnya install paket aplikasi DNS Servernya. Ketikkan perintah berikut :

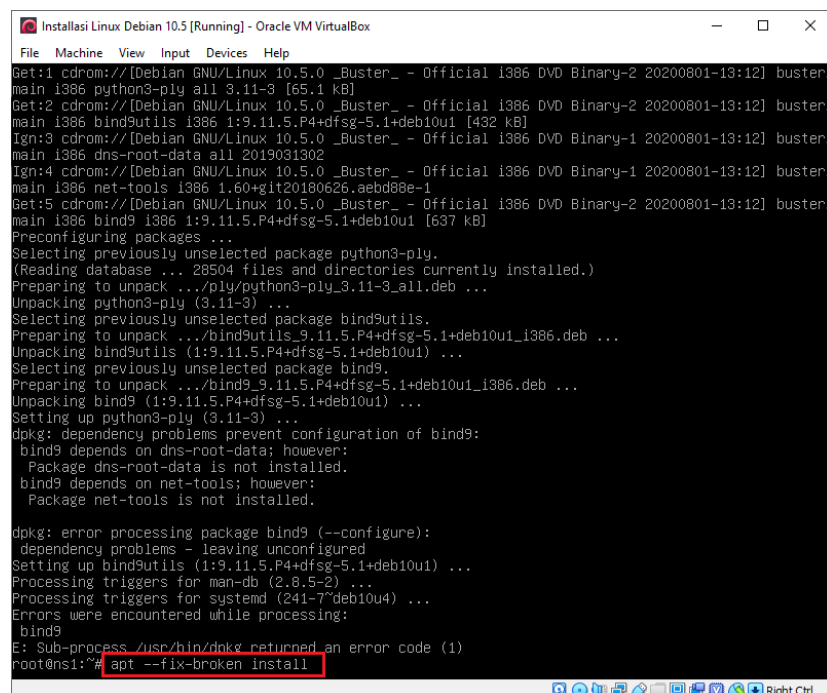
apt-get install -y bind9 >> *lalu tekan Enter*



```
tr@b main
Unmounting CD-ROM...
Repeat this process for the rest of the CDs in your set.
W: Failed to mount '/dev/sr0' to '/media/cdrom/'
root@ns1:~# apt-get update
Ign:1 cdrom://[Debian GNU/Linux 10.5.0 _Buster_ - Official i386 DVD Binary-2 20200801-13:12] buster
InRelease
Ign:2 cdrom://[Debian GNU/Linux 10.5.0 _Buster_ - Official i386 DVD Binary-3 20200801-13:12] buster
InRelease
Ign:3 cdrom://[Debian GNU/Linux 10.5.0 _Buster_ - Official i386 DVD Binary-1 20200801-13:12] buster
InRelease
Err:4 cdrom://[Debian GNU/Linux 10.5.0 _Buster_ - Official i386 DVD Binary-2 20200801-13:12] buster
Release
Please use apt-cdrom to make this CD-ROM recognized by APT. apt-get update cannot be used to add new CD-ROMs
Err:5 cdrom://[Debian GNU/Linux 10.5.0 _Buster_ - Official i386 DVD Binary-3 20200801-13:12] buster
Release
Please use apt-cdrom to make this CD-ROM recognized by APT. apt-get update cannot be used to add new CD-ROMs
Err:6 cdrom://[Debian GNU/Linux 10.5.0 _Buster_ - Official i386 DVD Binary-1 20200801-13:12] buster
Release
Please use apt-cdrom to make this CD-ROM recognized by APT. apt-get update cannot be used to add new CD-ROMs
Reading package lists... Done
E: The repository 'cdrom://[Debian GNU/Linux 10.5.0 _Buster_ - Official i386 DVD Binary-2 20200801-13:12] buster Release' does not have a Release file.
N: Updating from such a repository can't be done securely, and is therefore disabled by default.
N: See apt-secure(8) manpage for repository creation and user configuration details.
E: The repository 'cdrom://[Debian GNU/Linux 10.5.0 _Buster_ - Official i386 DVD Binary-3 20200801-13:12] buster Release' does not have a Release file.
N: Updating from such a repository can't be done securely, and is therefore disabled by default.
N: See apt-secure(8) manpage for repository creation and user configuration details.
E: The repository 'cdrom://[Debian GNU/Linux 10.5.0 _Buster_ - Official i386 DVD Binary-1 20200801-13:12] buster Release' does not have a Release file.
N: Updating from such a repository can't be done securely, and is therefore disabled by default.
N: See apt-secure(8) manpage for repository creation and user configuration details.
root@ns1:~# apt-get install -y bind9
```

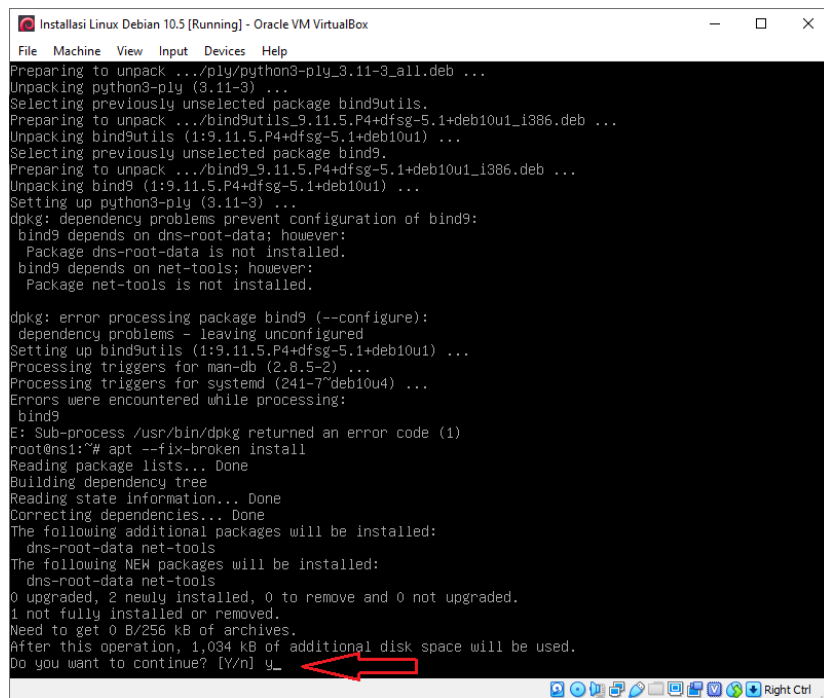
10. Masukkan perintah berikut untuk mengatasi broken pada paket DNS Server.

apt --fix-broken install >> *lalu tekan Enter*



```
Get:1 cdrom://[Debian GNU/Linux 10.5.0 _Buster_ - Official i386 DVD Binary-2 20200801-13:12] buster/
main i386 python3-ply all 3.11-3 [65.1 kB]
Get:2 cdrom://[Debian GNU/Linux 10.5.0 _Buster_ - Official i386 DVD Binary-2 20200801-13:12] buster/
main i386 bind9utils i386 1:9.11.5.P4+dfsg-5.1+deb10u1 [432 kB]
Ign:3 cdrom://[Debian GNU/Linux 10.5.0 _Buster_ - Official i386 DVD Binary-1 20200801-13:12] buster/
main i386 dns-root-data all 2019031302
Ign:4 cdrom://[Debian GNU/Linux 10.5.0 _Buster_ - Official i386 DVD Binary-1 20200801-13:12] buster/
main i386 net-tools i386 1.60+git20180626.aebd88e-1
Get:5 cdrom://[Debian GNU/Linux 10.5.0 _Buster_ - Official i386 DVD Binary-2 20200801-13:12] buster/
main i386 bind9 i386 1:9.11.5.P4+dfsg-5.1+deb10u1 [637 kB]
Preconfiguring packages ...
Selecting previously unselected package python3-ply.
(Reading database ... 28504 files and directories currently installed.)
Preparing to unpack .../ply/python3-ply_3.11-3_all.deb ...
Unpacking python3-ply (3.11-3) ...
Selecting previously unselected package bind9utils.
Preparing to unpack .../bind9utils_9.11.5.P4+dfsg-5.1+deb10u1_i386.deb ...
Unpacking bind9utils (1:9.11.5.P4+dfsg-5.1+deb10u1) ...
Selecting previously unselected package bind9.
Preparing to unpack .../bind9_9.11.5.P4+dfsg-5.1+deb10u1_i386.deb ...
Unpacking bind9 (1:9.11.5.P4+dfsg-5.1+deb10u1) ...
Setting up python3-ply (3.11-3) ...
dpkg: dependency problems prevent configuration of bind9:
 bind9 depends on dns-root-data; however:
  Package dns-root-data is not installed.
 bind9 depends on net-tools; however:
  Package net-tools is not installed.
dpkg: error processing package bind9 (--configure):
 dependency problems - leaving unconfigured
Setting up bind9utils (1:9.11.5.P4+dfsg-5.1+deb10u1) ...
Processing triggers for man-db (2.8.5-2) ...
Processing triggers for systemd (241-7~deb10u4) ...
Errors were encountered while processing:
 bind9
E: Sub-process /usr/bin/dpkg returned an error code (1)
root@ns1:~# apt --fix-broken install
```

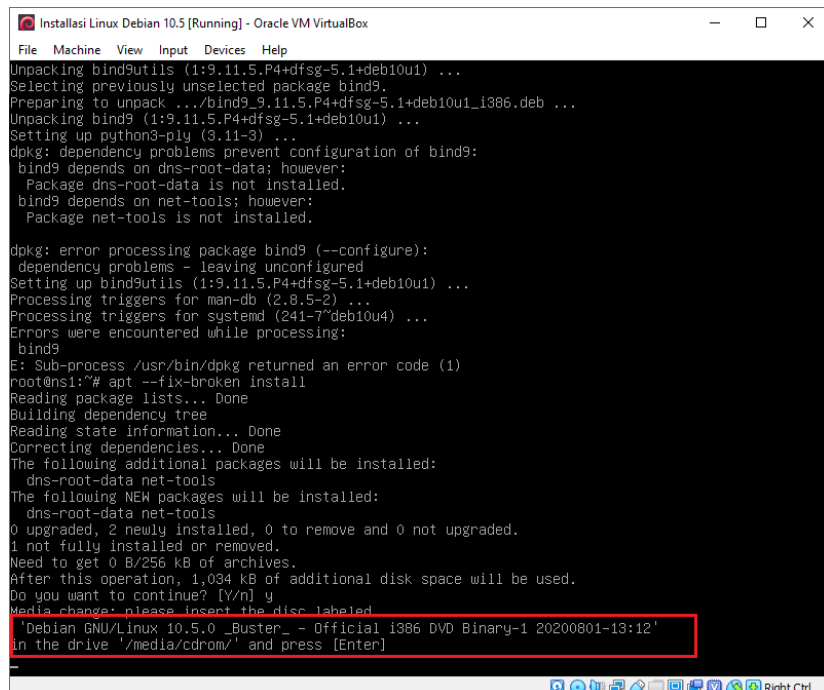

11. Jika muncul seperti gambar di bawah tekan tombol **Y** lalu Enter.



```
Installasi Linux Debian 10.5 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Preparing to unpack .../ply/python3-ply_3.11-3_all.deb ...
Unpacking python3-ply (3.11-3) ...
Selecting previously unselected package bind9utils.
Preparing to unpack .../bind9utils_9.11.5.P4+dfsg-5.1+deb10u1_i386.deb ...
Unpacking bind9utils (1:9.11.5.P4+dfsg-5.1+deb10u1) ...
Selecting previously unselected package bind9.
Preparing to unpack .../bind9_9.11.5.P4+dfsg-5.1+deb10u1_i386.deb ...
Unpacking bind9 (1:9.11.5.P4+dfsg-5.1+deb10u1) ...
Setting up python3-ply (3.11-3) ...
dpkg: dependency problems prevent configuration of bind9:
 bind9 depends on dns-root-data; however:
  Package dns-root-data is not installed.
 bind9 depends on net-tools; however:
  Package net-tools is not installed.

dpkg: error processing package bind9 (--configure):
 dependency problems - leaving unconfigured
Setting up bind9utils (1:9.11.5.P4+dfsg-5.1+deb10u1) ...
Processing triggers for man-db (2.8.5-2) ...
Processing triggers for systemd (241-7~deb10u4) ...
Errors were encountered while processing:
 bind9
E: Sub-process /usr/bin/dpkg returned an error code (1)
root@ns1:~# apt --fix-broken install
Reading package lists... Done
Building dependency tree
Reading state information... Done
Correcting dependencies... Done
The following additional packages will be installed:
 dns-root-data net-tools
The following NEW packages will be installed:
 dns-root-data net-tools
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
1 not fully installed or removed.
Need to get 0 B/256 kB of archives.
After this operation, 1,034 kB of additional disk space will be used.
Do you want to continue? [Y/n] y_
```

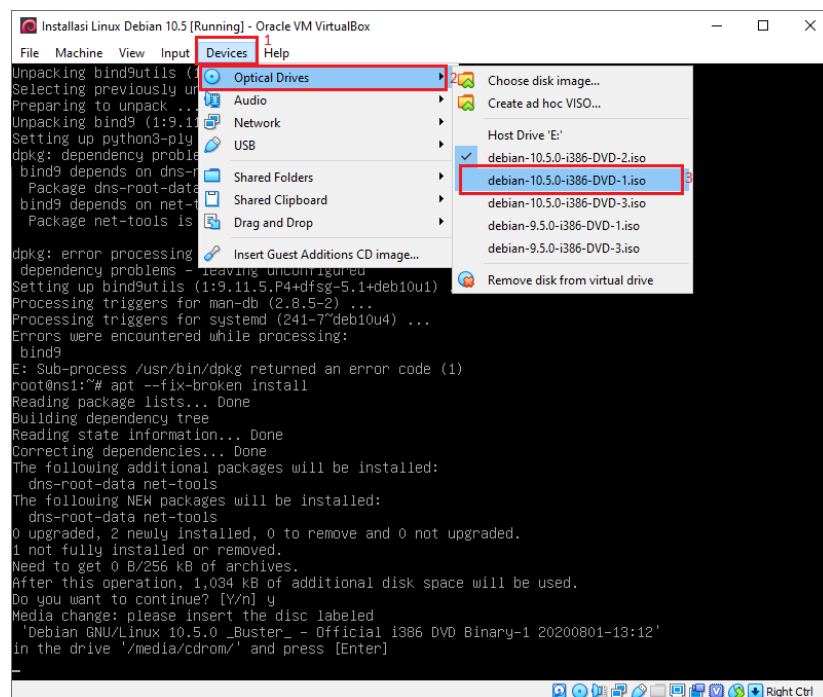
12. Jika tampil seperti gambar di bawah, jangan dulu di Enter. Masukkan terlebih dahulu DVD sesuai yang di minta. Masukkan DVD Binary-1 sesuai permintaan.



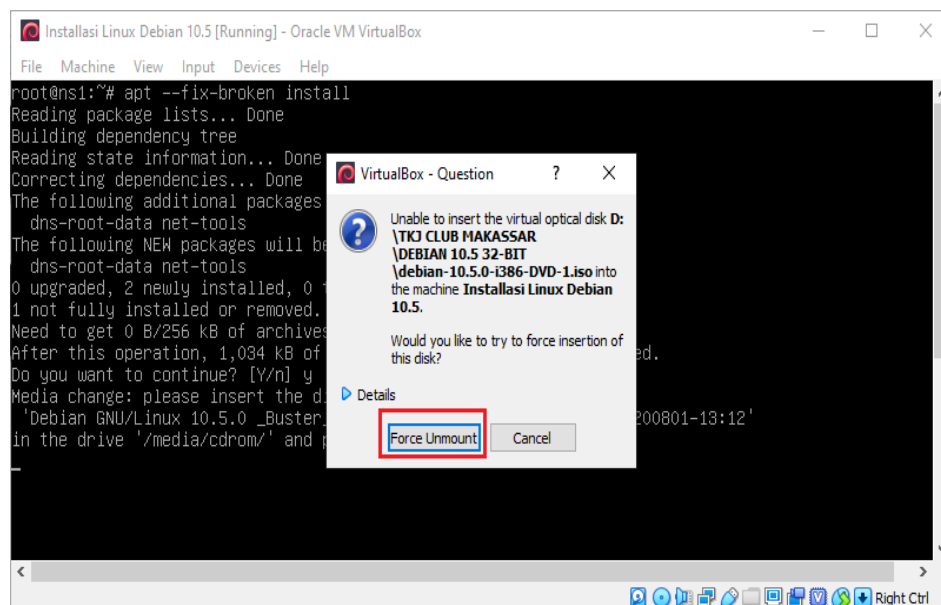
```
Installasi Linux Debian 10.5 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Unpacking bind9utils (1:9.11.5.P4+dfsg-5.1+deb10u1) ...
Selecting previously unselected package bind9.
Preparing to unpack .../bind9_9.11.5.P4+dfsg-5.1+deb10u1_i386.deb ...
Unpacking bind9 (1:9.11.5.P4+dfsg-5.1+deb10u1) ...
Setting up python3-ply (3.11-3) ...
dpkg: dependency problems prevent configuration of bind9:
 bind9 depends on dns-root-data; however:
  Package dns-root-data is not installed.
 bind9 depends on net-tools; however:
  Package net-tools is not installed.

dpkg: error processing package bind9 (--configure):
 dependency problems - leaving unconfigured
Setting up bind9utils (1:9.11.5.P4+dfsg-5.1+deb10u1) ...
Processing triggers for man-db (2.8.5-2) ...
Processing triggers for systemd (241-7~deb10u4) ...
Errors were encountered while processing:
 bind9
E: Sub-process /usr/bin/dpkg returned an error code (1)
root@ns1:~# apt --fix-broken install
Reading package lists... Done
Building dependency tree
Reading state information... Done
Correcting dependencies... Done
The following additional packages will be installed:
 dns-root-data net-tools
The following NEW packages will be installed:
 dns-root-data net-tools
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 0 B/256 kB of archives.
After this operation, 1,034 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Media change: please insert the disc labeled
''Debian GNU/Linux 10.5.0 _Buster_ - Official i386 DVD Binary-1 20200801-13:12''
in the drive '/media/cdrom/' and press [Enter]
```

13. Klik **Devices** >> **Optical Drives** >> lalu pilih **debian-10.5.0-i386-DVD-1.iso**.



14. Jika tampil seperti gambar di bawah. Klik **Force Unmount**.



15. Lalu tekan Enter.

```

root@ns1:~# apt --fix-broken install
Reading package lists... Done
Building dependency tree
Reading state information... Done
Correcting dependencies... Done
The following additional packages will be installed:
  dns-root-data net-tools
The following NEW packages will be installed:
  dns-root-data net-tools
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 0 B/256 kB of archives.
After this operation, 1,034 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Media change: please insert the disc labeled
'Debian GNU/Linux 10.5.0 _Buster_ - Official i386 DVD Binary-1 20200801-13:12'
in the drive '/media/cdrom/' and press [Enter]

```

16. Masukan perintah berikut, untuk menginstall paket **dnsutils** agar nantinya bisa menggunakan perintah **nslookup**.

apt-get install -y dnsutils

>> lalu tekan Enter

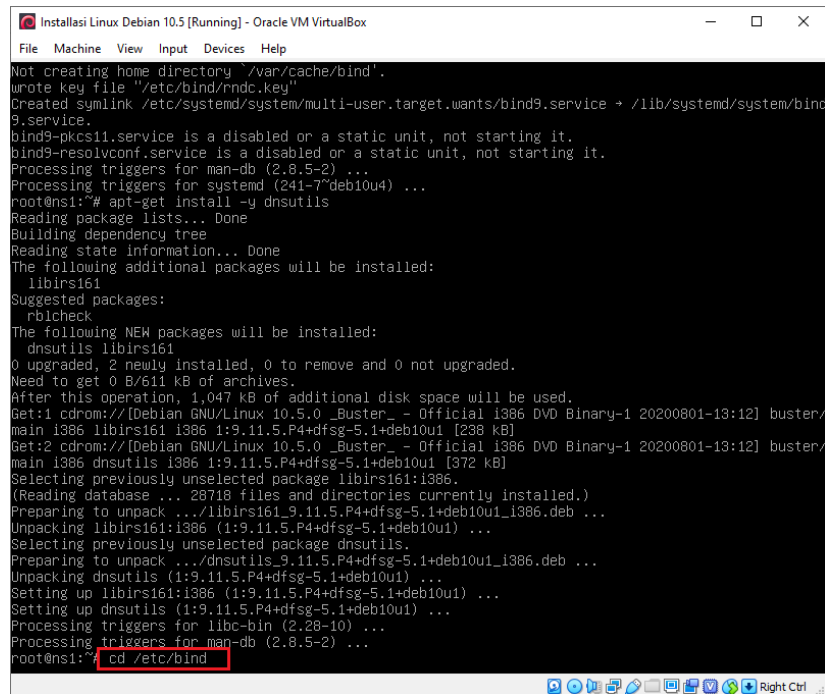
```

Get:1 cdrom://[Debian GNU/Linux 10.5.0 _Buster_ - Official i386 DVD Binary-1 20200801-13:12] buster/
main i386 dns-root-data all 2019031302 [5,396 B]
Get:2 cdrom://[Debian GNU/Linux 10.5.0 _Buster_ - Official i386 DVD Binary-1 20200801-13:12] buster/
main i386 net-tools i386 1.60+git20180626.aebd88e-1 [250 kB]
Selecting previously unselected package dns-root-data.
(Reading database ... 28653 files and directories currently installed.)
Preparing to unpack .../dns-root-data_2019031302_all.deb ...
Unpacking dns-root-data (2019031302) ...
Selecting previously unselected package net-tools.
Preparing to unpack .../net-tools_1.60+git20180626.aebd88e-1_i386.deb ...
Unpacking net-tools (1.60+git20180626.aebd88e-1) ...
Setting up net-tools (1.60+git20180626.aebd88e-1) ...
Setting up dns-root-data (2019031302) ...
Setting up bind9 (1:9.11.5.P4+dfsg-5.1+deb10u1) ...
Adding group 'bind' (GID 114) ...
Done.
Adding system user 'bind' (UID 107) ...
Adding new user 'bind' (UID 107) with group 'bind' ...
Not creating home directory '/var/cache/bind'.
wrote key file "/etc/bind/rndc.key"
Created symlink /etc/systemd/system/multi-user.target.wants/bind9.service + /lib/systemd/system/bind
9.service.
bind9-pkcs11.service is a disabled or a static unit, not starting it.
bind9-resolvconf.service is a disabled or a static unit, not starting it.
Processing triggers for man-db (2.8.5-2) ...
Processing triggers for systemd (241-7deb10u4) ...
root@ns1:~# apt-get install -y dnsutils

```

17. Setelah selesai menginstall paketnya, Langkah selanjutnya yaitu pindah ke directory **bind** dengan cara masukkan perintah berikut :

```
# cd /etc/bind >> lalu tekan Enter
```

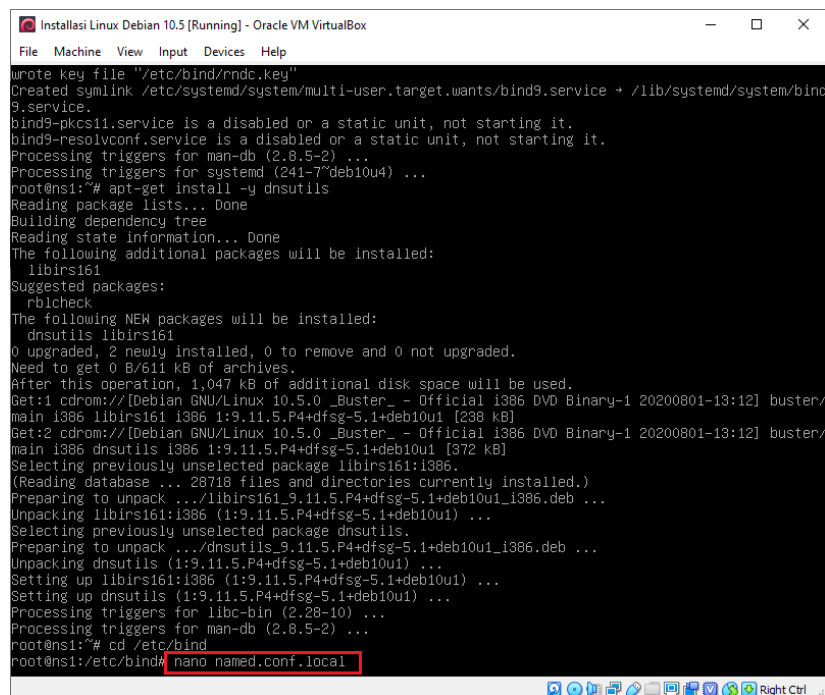


```
Instalasi Linux Debian 10.5 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

Not creating home directory "/var/cache/bind".
wrote key file "/etc/bind/rndc.key"
Created symlink /etc/systemd/system/multi-user.target.wants/bind9.service → /lib/systemd/system/bind9.service.
bind9-pkcs11.service is a disabled or a static unit, not starting it.
bind9-resolvconf.service is a disabled or a static unit, not starting it.
Processing triggers for man-db (2.8.5-2) ...
Processing triggers for systemd (241-7~deb10u4) ...
root@ns1:~# apt-get install -y dnsmasq
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  libnss3
Suggested packages:
  rblcheck
The following NEW packages will be installed:
  dnsmasq libnss3
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 0 B/611 kB of archives.
After this operation, 1,047 kB of additional disk space will be used.
Get:1 cdrom://[Debian GNU/Linux 10.5.0 _Buster_ - Official i386 DVD Binary-1 20200801-13:12] buster/main i386 libnss3 i386 1:9.11.5.P4+dfsg-5.1+deb10u1 [238 kB]
Get:2 cdrom://[Debian GNU/Linux 10.5.0 _Buster_ - Official i386 DVD Binary-1 20200801-13:12] buster/main i386 dnsmasq i386 1:9.11.5.P4+dfsg-5.1+deb10u1 [372 kB]
Selecting previously unselected package libnss3:i386.
(Reading database ... 28718 files and directories currently installed.)
Preparing to unpack .../libnss3_9.11.5.P4+dfsg-5.1+deb10u1_i386.deb ...
Unpacking libnss3:i386 (1:9.11.5.P4+dfsg-5.1+deb10u1) ...
Selecting previously unselected package dnsmasq.
Preparing to unpack .../dnsmasq_9.11.5.P4+dfsg-5.1+deb10u1_i386.deb ...
Unpacking dnsmasq (1:9.11.5.P4+dfsg-5.1+deb10u1) ...
Setting up libnss3:i386 (1:9.11.5.P4+dfsg-5.1+deb10u1) ...
Setting up dnsmasq (1:9.11.5.P4+dfsg-5.1+deb10u1) ...
Processing triggers for libc-bin (2.28-10) ...
Processing triggers for man-db (2.8.5-2) ...
root@ns1:~# cd /etc/bind
```

18. Masukkan perintah berikut untuk mengedit file **named.conf.local**.

```
# nano named.conf.local >> lalu tekan Enter
```



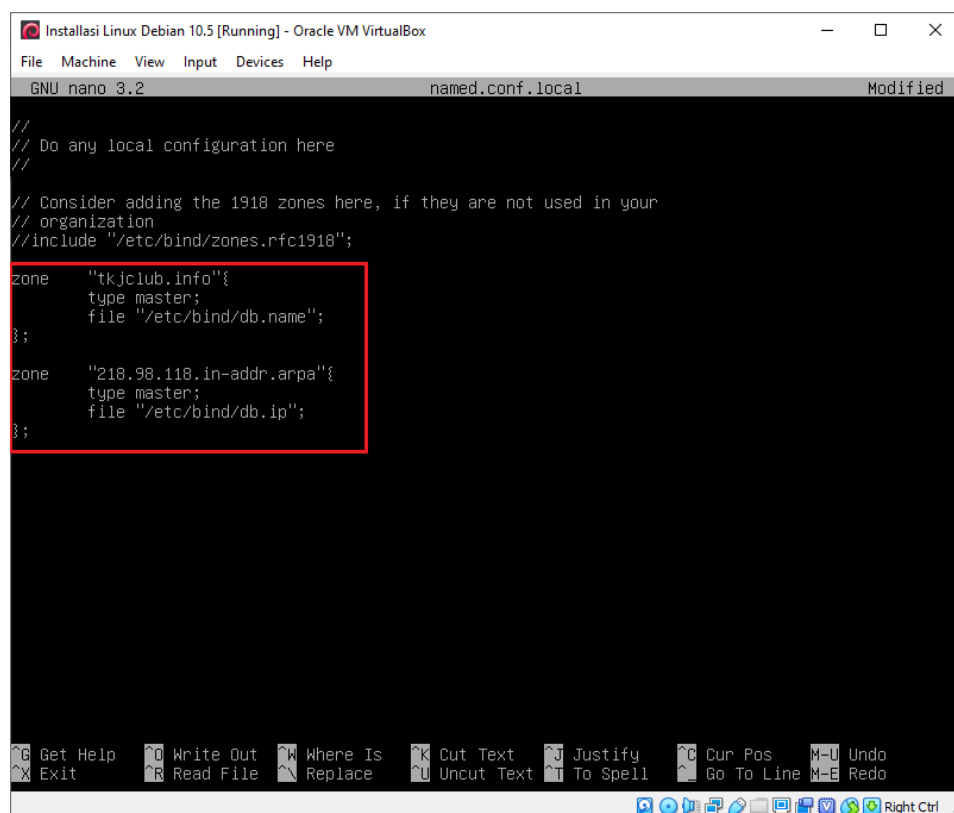
```
Instalasi Linux Debian 10.5 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

wrote key file "/etc/bind/rndc.key"
Created symlink /etc/systemd/system/multi-user.target.wants/bind9.service → /lib/systemd/system/bind9.service.
bind9-pkcs11.service is a disabled or a static unit, not starting it.
bind9-resolvconf.service is a disabled or a static unit, not starting it.
Processing triggers for man-db (2.8.5-2) ...
Processing triggers for systemd (241-7~deb10u4) ...
root@ns1:~# apt-get install -y dnsmasq
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  libnss3
Suggested packages:
  rblcheck
The following NEW packages will be installed:
  dnsmasq libnss3
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 0 B/611 kB of archives.
After this operation, 1,047 kB of additional disk space will be used.
Get:1 cdrom://[Debian GNU/Linux 10.5.0 _Buster_ - Official i386 DVD Binary-1 20200801-13:12] buster/main i386 libnss3 i386 1:9.11.5.P4+dfsg-5.1+deb10u1 [238 kB]
Get:2 cdrom://[Debian GNU/Linux 10.5.0 _Buster_ - Official i386 DVD Binary-1 20200801-13:12] buster/main i386 dnsmasq i386 1:9.11.5.P4+dfsg-5.1+deb10u1 [372 kB]
Selecting previously unselected package libnss3:i386.
(Reading database ... 28718 files and directories currently installed.)
Preparing to unpack .../libnss3_9.11.5.P4+dfsg-5.1+deb10u1_i386.deb ...
Unpacking libnss3:i386 (1:9.11.5.P4+dfsg-5.1+deb10u1) ...
Selecting previously unselected package dnsmasq.
Preparing to unpack .../dnsmasq_9.11.5.P4+dfsg-5.1+deb10u1_i386.deb ...
Unpacking dnsmasq (1:9.11.5.P4+dfsg-5.1+deb10u1) ...
Setting up libnss3:i386 (1:9.11.5.P4+dfsg-5.1+deb10u1) ...
Setting up dnsmasq (1:9.11.5.P4+dfsg-5.1+deb10u1) ...
Processing triggers for libc-bin (2.28-10) ...
Processing triggers for man-db (2.8.5-2) ...
root@ns1:~# cd /etc/bind
root@ns1:/etc/bind# nano named.conf.local
```

19. Maka akan tampil seperti gambar berikut. Lalu tambahkan script di bagian paling bawah seperti gambar di bawah.

```
zone "tkjclub.info"{
    type master;
    file "/etc/bind/db.name";
};

zone "218.98.118.in-addr.arpa"{
    type master;
    file "/etc/bind/db.ip";
};
```



```
Instalasi Linux Debian 10.5 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
GNU nano 3.2 named.conf.local Modified

//
// Do any local configuration here
//
// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";

zone "tkjclub.info"{
    type master;
    file "/etc/bind/db.name";
};

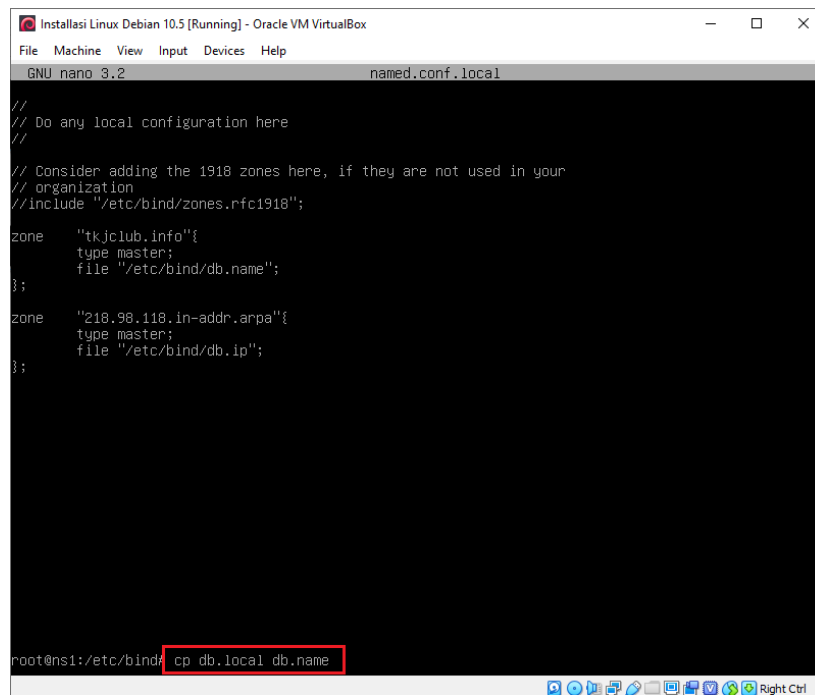
zone "218.98.118.in-addr.arpa"{
    type master;
    file "/etc/bind/db.ip";
};

^G Get Help  ^O Write Out  ^W Where Is   ^K Cut Text   ^J Justify    ^C Cur Pos    M-U Undo
^X Exit      ^R Read File  ^N Replace    ^U Uncut Text ^T To Spell   ^G Go To Line M-E Redo
Right Ctrl
```

20. Lalu save dengan cara menekan tombol kombinasi **Ctrl + X** lalu tekan tombol **Y** kemudian **Enter**.

21. Copy file **db.local** ke file **db.name**. Masukkan perintah berikut :

cp db.local db.name >> *lalu tekan Enter*



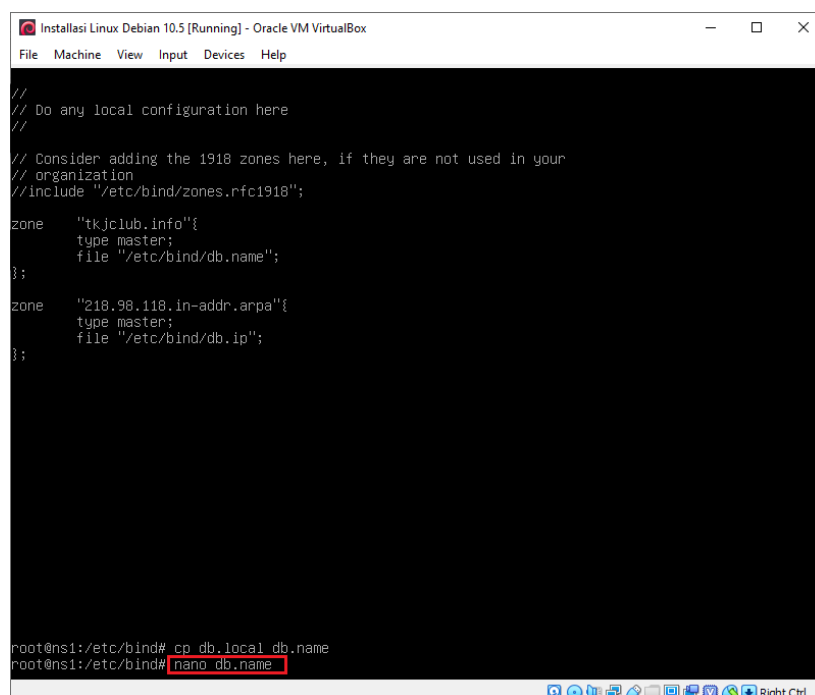
The screenshot shows a terminal window titled "Instalasi Linux Debian 10.5 [Running] - Oracle VM VirtualBox". The window contains the output of the 'cat' command for the file '/etc/bind/named.conf.local'. The content of the file is as follows:

```
//  
// Do any local configuration here  
//  
// Consider adding the 1918 zones here, if they are not used in your  
// organization  
//include "/etc/bind/zones.rfc1918";  
  
zone "tkjclub.info"{  
    type master;  
    file "/etc/bind/db.name";  
};  
  
zone "218.98.118.in-addr.arpa"{  
    type master;  
    file "/etc/bind/db.ip";  
};
```

At the bottom of the terminal, the command `root@ns1:/etc/bind# cp db.local db.name` is being entered, with the command text highlighted by a red box.

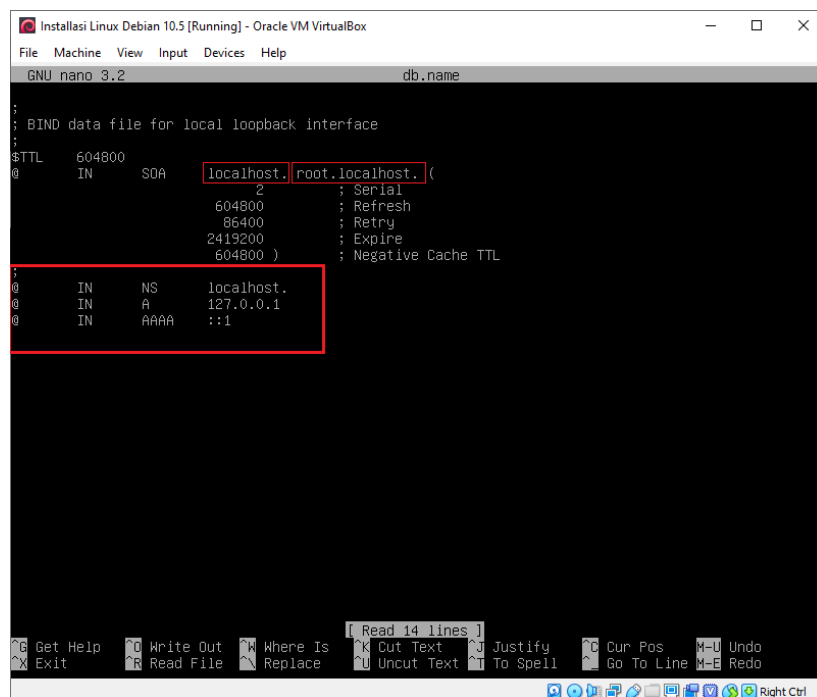
22. Kemudian edit file **db.name**. Ketikkan perintah berikut :

nano db.name >> *lalu tekan Enter*



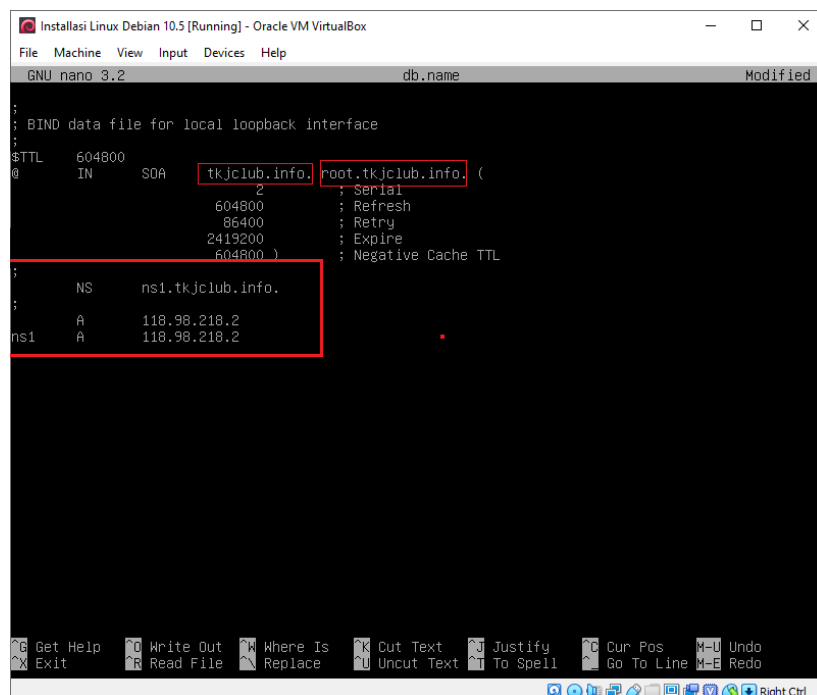
The screenshot shows the same terminal window as before. The command `root@ns1:/etc/bind# cp db.local db.name` has been executed. The next command, `root@ns1:/etc/bind# nano db.name`, is being entered, with the command text highlighted by a red box.

23. Maka akan tampil seperti gambar di bawah, lalu edit scrip di bawah .



```
GNU nano 3.2 db.name
; BIND data file for local loopback interface
$TTL 604800
@ IN SOA localhost. root.localhost. (
    2 ; Serial
    604800 ; Refresh
    86400 ; Retry
    2419200 ; Expire
    604800 ) ; Negative Cache TTL
;
@ IN NS localhost.
@ IN A 127.0.0.1
@ IN AAAA ::1
```

24. Edit script menjadi seperti gambar berikut.

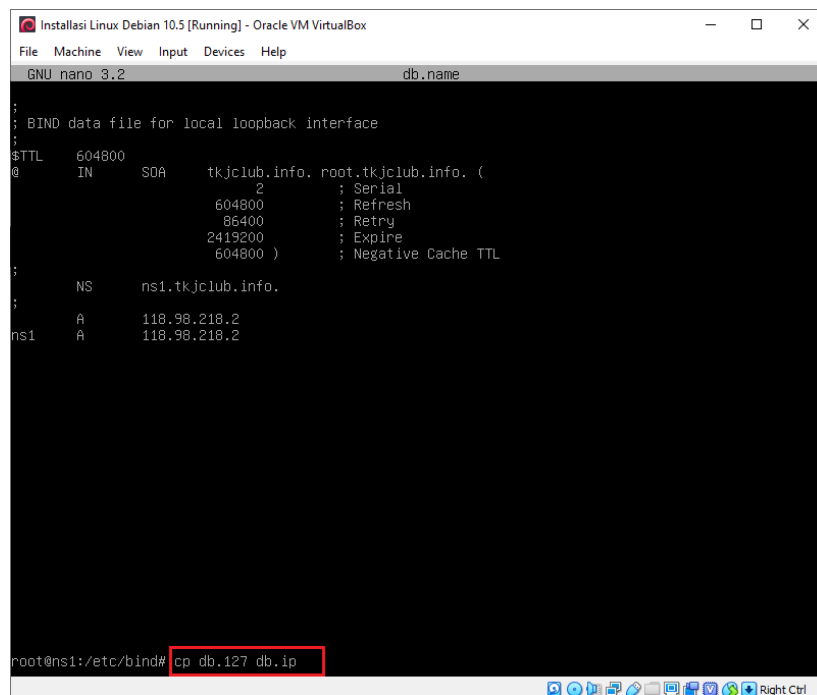


```
GNU nano 3.2 db.name Modified
; BIND data file for local loopback interface
$TTL 604800
@ IN SOA tkjclub.info. root.tkjclub.info. (
    2 ; Serial
    604800 ; Refresh
    86400 ; Retry
    2419200 ; Expire
    604800 ) ; Negative Cache TTL
;
@ IN NS ns1.tkjclub.info.
@ IN A 118.98.218.2
@ IN AAAA ::1
```

25. Lalu save dengan cara menekan tombol kombinasi **Ctrl + X** lalu tekan tombol **Y** kemudian **Enter**.

26. Copy file **db.127** ke file **db.ip**. Masukkan perintah berikut :

cp db.127 db.ip >> *lalu tekan Enter*

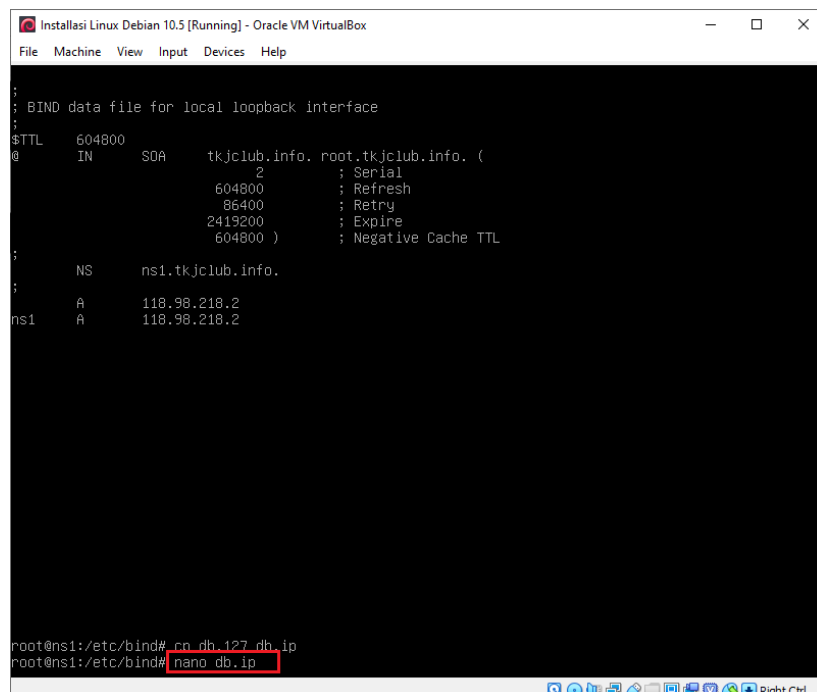


```
Installasi Linux Debian 10.5 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
GNU nano 3.2 db.name
;
; BIND data file for local loopback interface
;
$TTL 604800
@ IN SOA tkjclub.info. root.tkjclub.info. (
    2      ; Serial
    604800 ; Refresh
    86400  ; Retry
    2419200 ; Expire
    604800 ) ; Negative Cache TTL
;
NS ns1.tkjclub.info.
;
A 118.98.218.2
ns1 A 118.98.218.2

root@ns1:/etc/bind# cp db.127 db.ip
```

27. Kemudian edit file **db.ip**. Ketikkan perintah berikut :

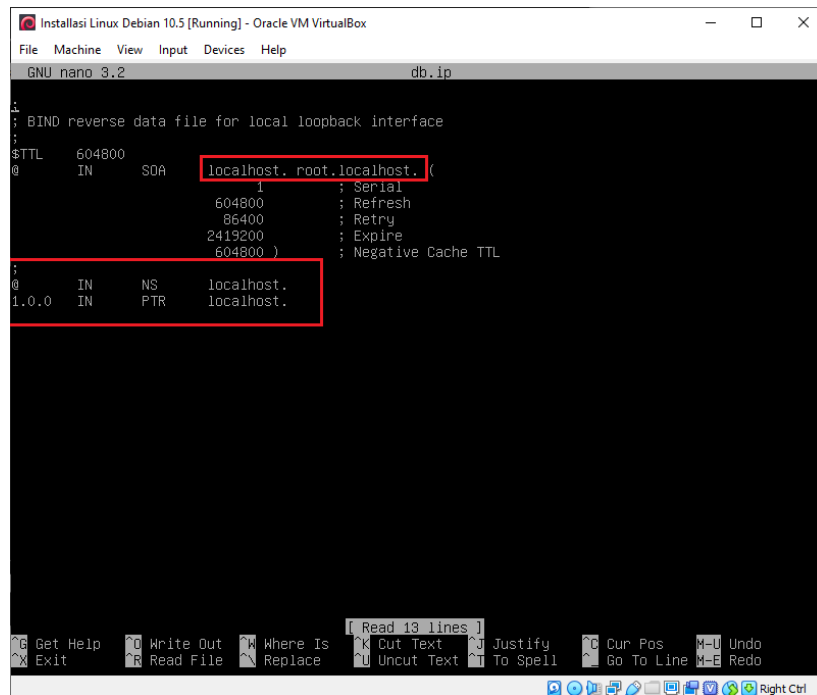
nano db.ip >> *lalu tekan Enter*



```
Installasi Linux Debian 10.5 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
;
; BIND data file for local loopback interface
;
$TTL 604800
@ IN SOA tkjclub.info. root.tkjclub.info. (
    2      ; Serial
    604800 ; Refresh
    86400  ; Retry
    2419200 ; Expire
    604800 ) ; Negative Cache TTL
;
NS ns1.tkjclub.info.
;
A 118.98.218.2
ns1 A 118.98.218.2

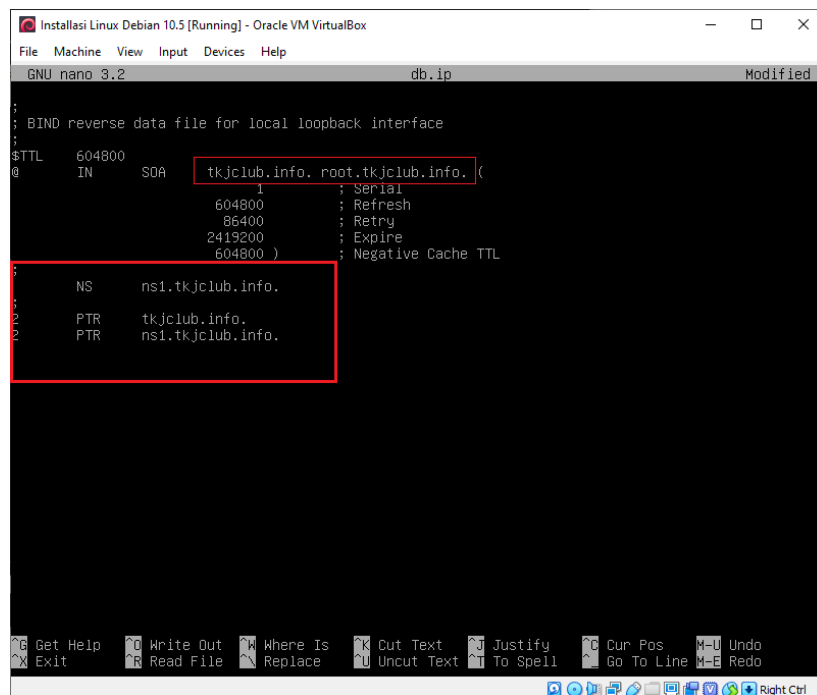
root@ns1:/etc/bind# cp db.127 db.ip
root@ns1:/etc/bind# nano db.ip
```


28. Maka akan tampil seperti gambar di bawah, lalu edit scrip di bawah.



```
GNU nano 3.2 db.ip
; BIND reverse data file for local loopback interface
$TTL 604800
@ IN SOA localhost. root.localhost. (
    1          ; Serial
    604800     ; Refresh
    86400      ; Retry
    2419200    ; Expire
    604800 )   ; Negative Cache TTL
;
@ IN NS localhost.
1.0.0 IN PTR localhost.
```

29. Edit script menjadi seperti gambar berikut.

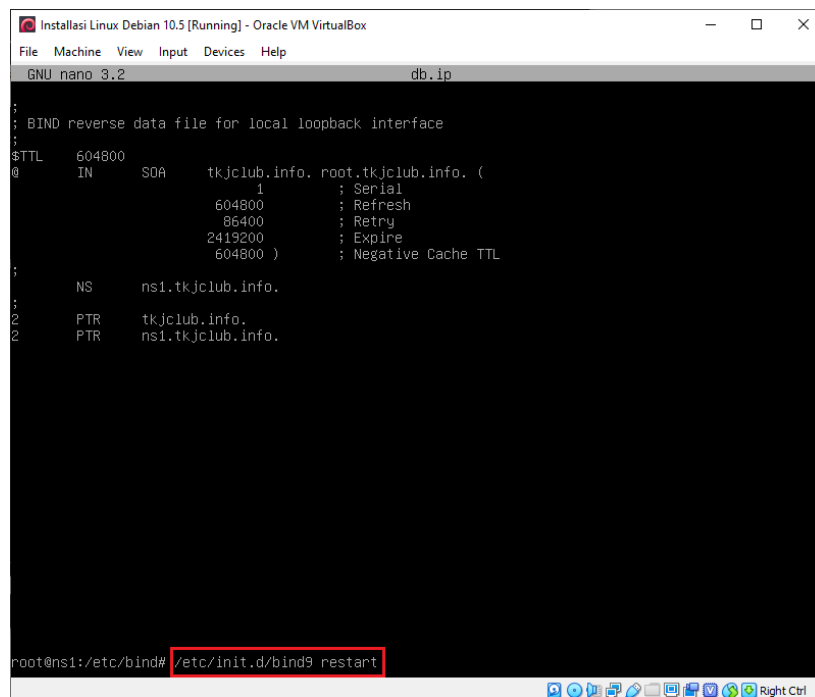


```
GNU nano 3.2 db.ip Modified
; BIND reverse data file for local loopback interface
$TTL 604800
@ IN SOA tkjclub.info. root.tkjclub.info. (
    1          ; Serial
    604800     ; Refresh
    86400      ; Retry
    2419200    ; Expire
    604800 )   ; Negative Cache TTL
;
NS ns1.tkjclub.info.
PTR tkjclub.info.
PTR ns1.tkjclub.info.
```

30. Lalu save dengan cara menekan tombol kombinasi **Ctrl + X** lalu tekan tombol **Y** kemudian **Enter**.

31. Selanjutnya, restart **bind9** menggunakan perintah berikut :

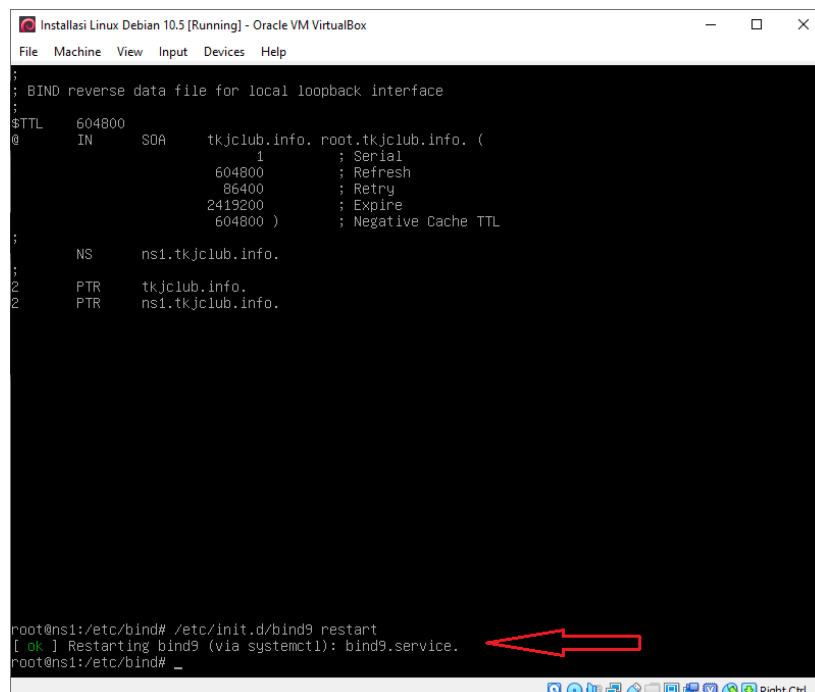
/etc/init.d/bind9 restart >> *lalu tekan Enter*



```
GNU nano 3.2 db.ip
; BIND reverse data file for local loopback interface
$TTL 604800
@ IN SOA tkjclub.info. root.tkjclub.info. (
    1 ; Serial
    604800 ; Refresh
    86400 ; Retry
    2419200 ; Expire
    604800 ) ; Negative Cache TTL
;
NS ns1.tkjclub.info.
PTR tkjclub.info.
PTR ns1.tkjclub.info.

root@ns1:/etc/bind# /etc/init.d/bind9 restart
```

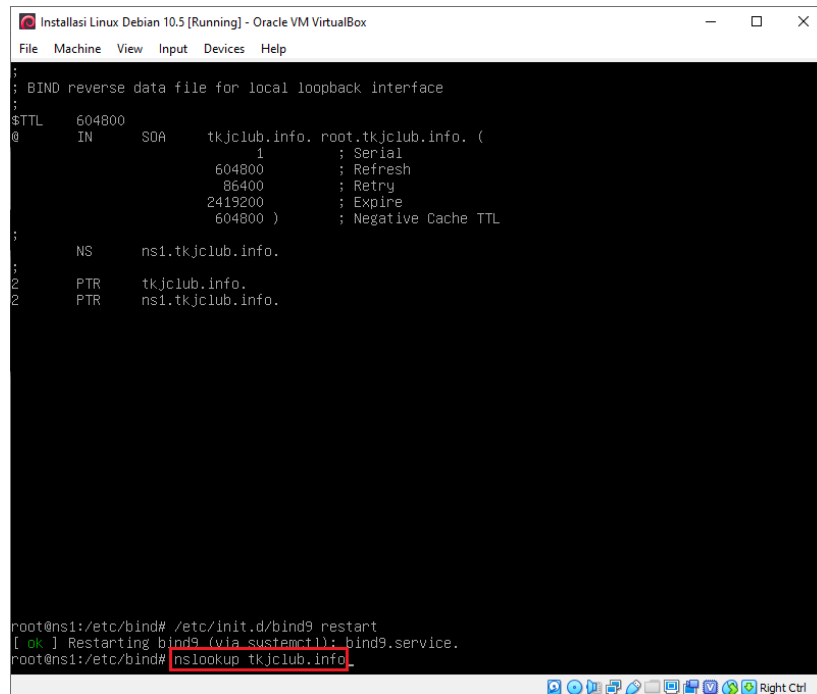
32. Pastikan hasilnya **OK**.



```
root@ns1:/etc/bind# /etc/init.d/bind9 restart
[ ok ] Restarting bind9 (via systemctl): bind9.service.
root@ns1:/etc/bind#
```

33. Setelah itu, lakukan pengecekan pada nama domain yang di buat apakah sudah berhasil. Menggunakan perintah **nslookup**.

#nslookup tkjclub.info >> *lalu tekan Enter*

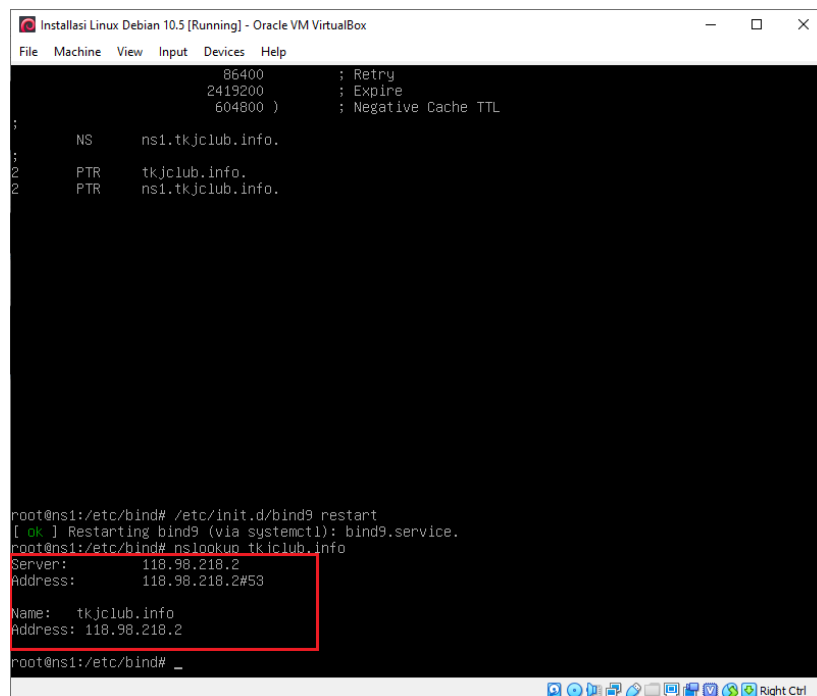


```
Instalasi Linux Debian 10.5 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

; BIND reverse data file for local loopback interface
$TTL 604800
@ IN SOA tkjclub.info. root.tkjclub.info. (
    1 ; Serial
    604800 ; Refresh
    86400 ; Retry
    2419200 ; Expire
    604800 ) ; Negative Cache TTL
;
NS ns1.tkjclub.info.
PTR tkjclub.info.
PTR ns1.tkjclub.info.

root@ns1:/etc/bind# /etc/init.d/bind9 restart
[ ok ] Restarting bind9 (via systemctl): bind9.service.
root@ns1:/etc/bind# nslookup tkjclub.info
```

34. Jika berhasil maka hasilnya seperti tampilan gambar di bawah.



```
Instalasi Linux Debian 10.5 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

; BIND reverse data file for local loopback interface
$TTL 604800
@ IN SOA tkjclub.info. root.tkjclub.info. (
    1 ; Serial
    604800 ; Refresh
    86400 ; Retry
    2419200 ; Expire
    604800 ) ; Negative Cache TTL
;
NS ns1.tkjclub.info.
PTR tkjclub.info.
PTR ns1.tkjclub.info.

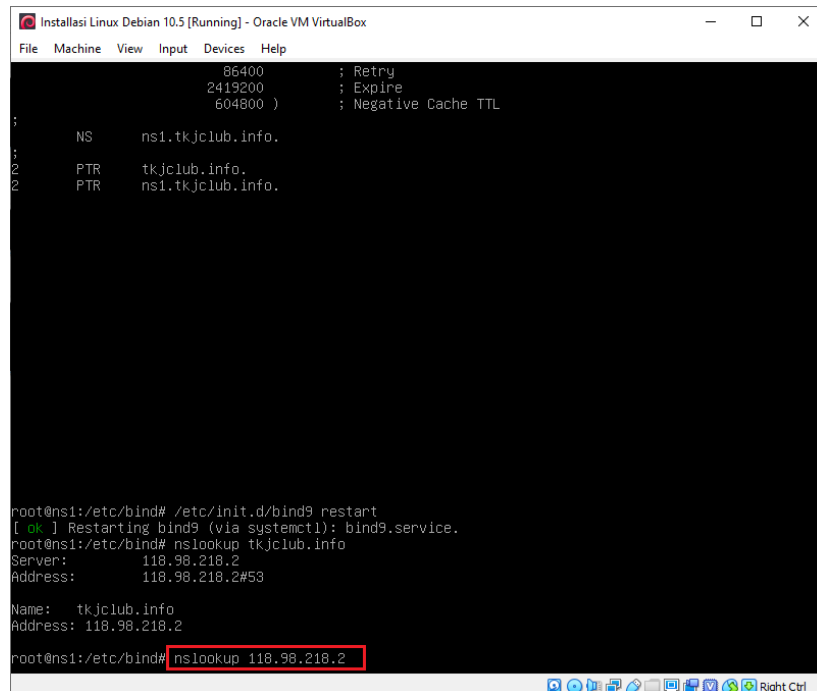
root@ns1:/etc/bind# /etc/init.d/bind9 restart
[ ok ] Restarting bind9 (via systemctl): bind9.service.
root@ns1:/etc/bind# nslookup tkjclub.info
Server: 118.98.218.2
Address: 118.98.218.2#53

Name: tkjclub.info
Address: 118.98.218.2

root@ns1:/etc/bind#
```

35. Lakukan pengecekan pada alamat IP yang di arahkan dalam pembuatan nama domain apakah sudah berhasil. Menggunakan perintah **nslookup**

#nslookup 118.98.218.2 >> lalu tekan Enter



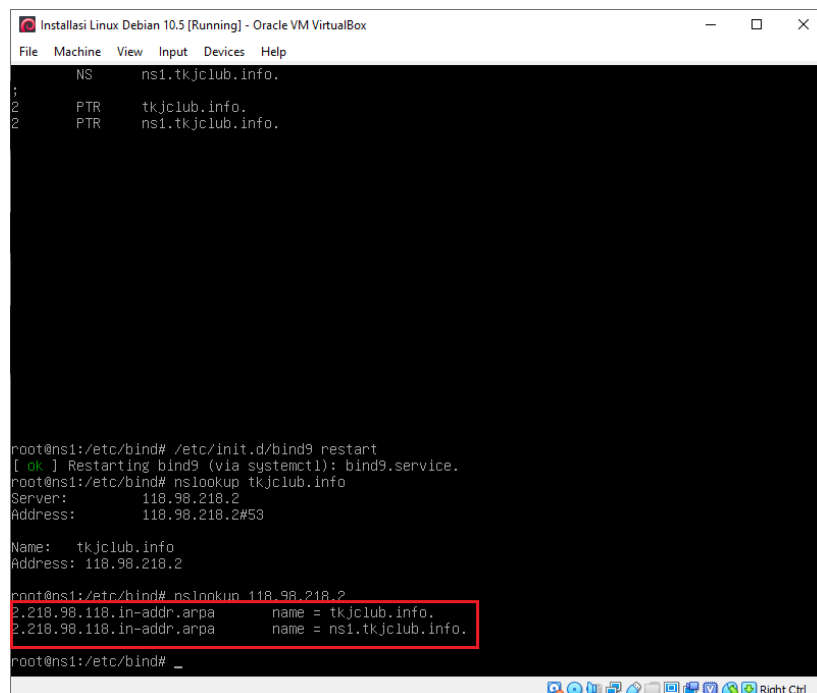
```
Instalasi Linux Debian 10.5 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

      86400      ; Retry
     2419200    ; Expire
    604800 )    ; Negative Cache TTL
;
      NS       ns1.tkjclub.info.
;
      PTR      tkjclub.info.
      PTR      ns1.tkjclub.info.
;

root@ns1:/etc/bind# /etc/init.d/bind9 restart
[ ok ] Restarting bind9 (via systemctl): bind9.service.
root@ns1:/etc/bind# nslookup tkjclub.info
Server:      118.98.218.2
Address:     118.98.218.2#53

Name:   tkjclub.info
Address: 118.98.218.2
root@ns1:/etc/bind# nslookup 118.98.218.2
```

36. Jika berhasil maka hasilnya seperti tampilan gambar di bawah.



```
Instalasi Linux Debian 10.5 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

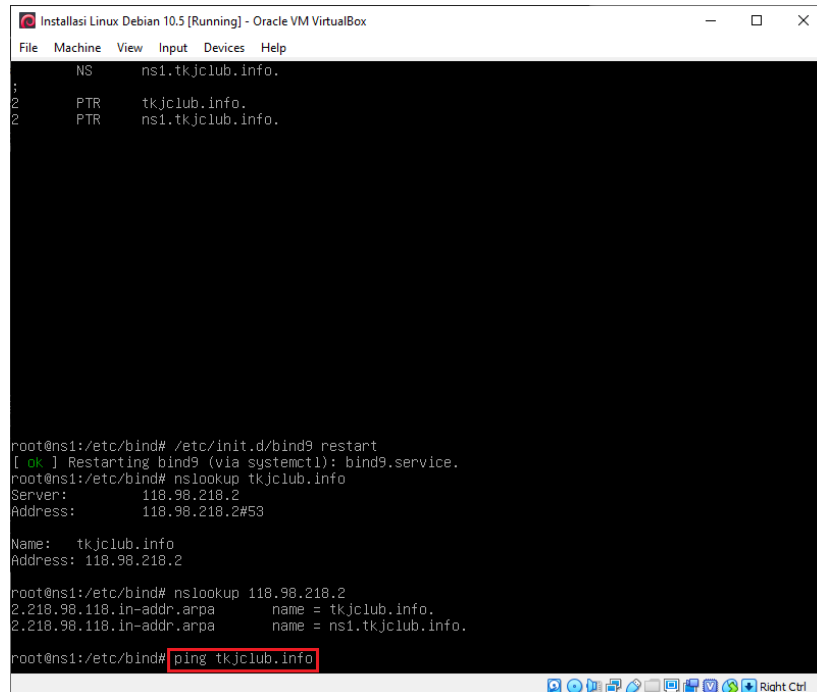
      NS       ns1.tkjclub.info.
;
      PTR      tkjclub.info.
      PTR      ns1.tkjclub.info.
;

root@ns1:/etc/bind# /etc/init.d/bind9 restart
[ ok ] Restarting bind9 (via systemctl): bind9.service.
root@ns1:/etc/bind# nslookup tkjclub.info
Server:      118.98.218.2
Address:     118.98.218.2#53

Name:   tkjclub.info
Address: 118.98.218.2
root@ns1:/etc/bind# nslookup 118.98.218.2
2.218.98.118.in-addr.arpa    name = tkjclub.info.
2.218.98.118.in-addr.arpa    name = ns1.tkjclub.info.
root@ns1:/etc/bind# _
```

37. Lakukan lagi pengecekan pada nama domain apakah sudah berhasil. menggunakan perintah **ping**.

ping tkjclub.info >>lalu tekan *Enter*



```
Instalasi Linux Debian 10.5 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

NS      ns1.tkjclub.info.
;
2 PTR   tkjclub.info.
2 PTR   ns1.tkjclub.info.

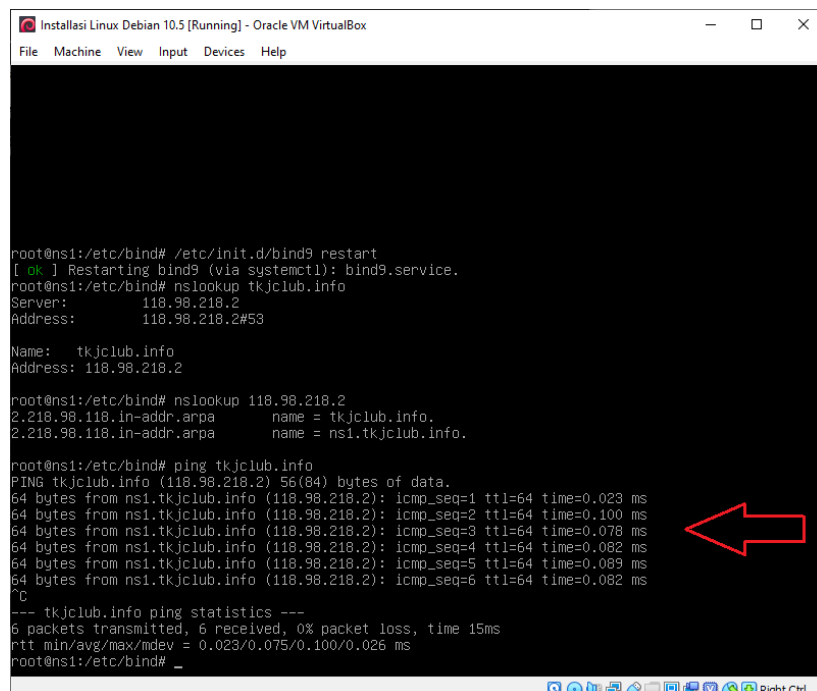
root@ns1:/etc/bind# /etc/init.d/bind9 restart
[ ok ] Restarting bind9 (via systemctl): bind9.service.
root@ns1:/etc/bind# nslookup tkjclub.info
Server:      118.98.218.2
Address:     118.98.218.2#53

Name:   tkjclub.info
Address: 118.98.218.2

root@ns1:/etc/bind# nslookup 118.98.218.2
2.218.98.118.in-addr.arpa    name = tkjclub.info.
2.218.98.118.in-addr.arpa    name = ns1.tkjclub.info.

root@ns1:/etc/bind# ping tkjclub.info
```

38. Pastikan hasilnya seperti gambar di bawah. “untuk menghentikan proses ping, tekan tombol **ctrl + c**”.



```
Instalasi Linux Debian 10.5 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

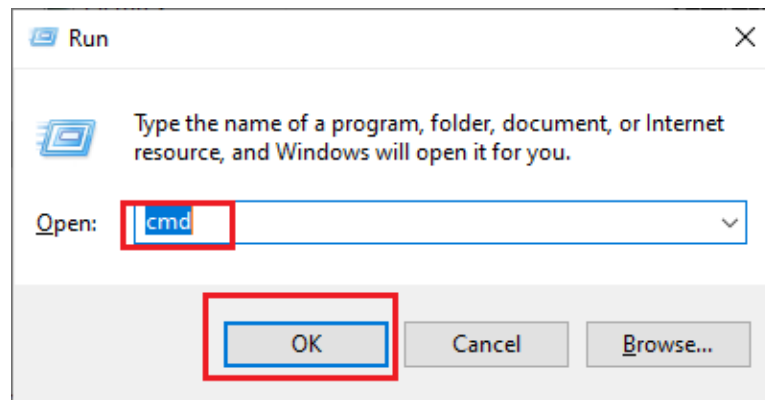
root@ns1:/etc/bind# /etc/init.d/bind9 restart
[ ok ] Restarting bind9 (via systemctl): bind9.service.
root@ns1:/etc/bind# nslookup tkjclub.info
Server:      118.98.218.2
Address:     118.98.218.2#53

Name:   tkjclub.info
Address: 118.98.218.2

root@ns1:/etc/bind# nslookup 118.98.218.2
2.218.98.118.in-addr.arpa    name = tkjclub.info.
2.218.98.118.in-addr.arpa    name = ns1.tkjclub.info.

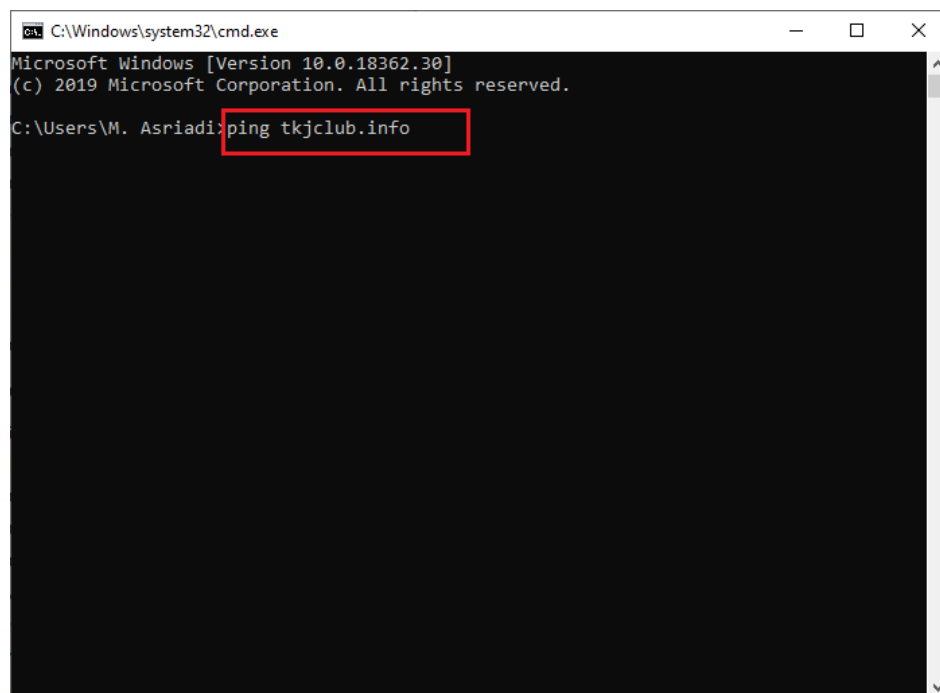
root@ns1:/etc/bind# ping tkjclub.info
PING tkjclub.info (118.98.218.2) 56(84) bytes of data:
64 bytes from ns1.tkjclub.info (118.98.218.2): icmp_seq=1 ttl=64 time=0.023 ms
64 bytes from ns1.tkjclub.info (118.98.218.2): icmp_seq=2 ttl=64 time=0.100 ms
64 bytes from ns1.tkjclub.info (118.98.218.2): icmp_seq=3 ttl=64 time=0.078 ms
64 bytes from ns1.tkjclub.info (118.98.218.2): icmp_seq=4 ttl=64 time=0.082 ms
64 bytes from ns1.tkjclub.info (118.98.218.2): icmp_seq=5 ttl=64 time=0.089 ms
64 bytes from ns1.tkjclub.info (118.98.218.2): icmp_seq=6 ttl=64 time=0.082 ms
^C
--- tkjclub.info ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 15ms
rtt min/avg/max/mdev = 0.023/0.075/0.100/0.026 ms
root@ns1:/etc/bind# _
```

39. Untuk lebih memastikan apakah konfigurasi **DNS Server** telah selesai dan berhasil, lakukan test **ping** di Client (Windows). Tekan tombol kombinasi **Windows + R** (Run), kemudian ketikkan **CMD** lalu klik **OK**.

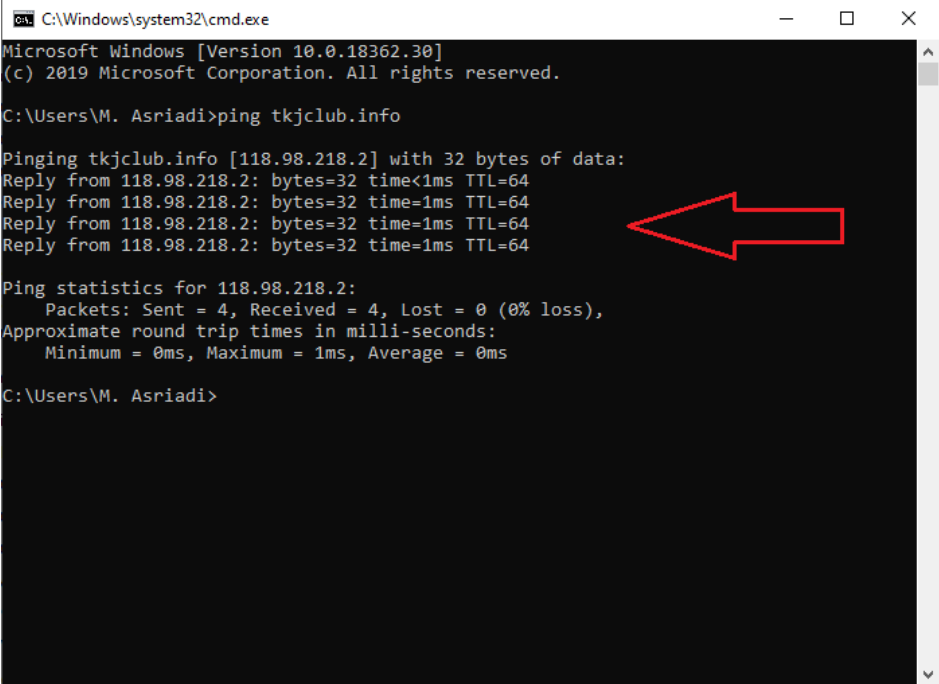


40. Jika **CMD** sudah terbuka, lakukan tes ping nama domain. ketikkan perintah berikut :

ping tkjclub.info >> *lalu tekan Enter*



41. Pastikan hasilnya **Reply** seperti pada gambar di bawah.



```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.18362.30]
(c) 2019 Microsoft Corporation. All rights reserved.

C:\Users\M. Asriadi>ping tkjclub.info

Pinging tkjclub.info [118.98.218.2] with 32 bytes of data:
Reply from 118.98.218.2: bytes=32 time<1ms TTL=64
Reply from 118.98.218.2: bytes=32 time=1ms TTL=64
Reply from 118.98.218.2: bytes=32 time=1ms TTL=64
Reply from 118.98.218.2: bytes=32 time=1ms TTL=64

Ping statistics for 118.98.218.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\M. Asriadi>
```

A red arrow points to the four 'Reply' lines in the ping output.

42. Jika seluruh step di atas sudah di lakukan, maka konfigurasi DNS Server telah selesai dan berhasil.

****SALAM SUKSES TKJ CLUB****